

Bera Yavuz

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EDUCATION

Bachelor of Applied Science in Nanotechnology Engineering

University of Waterloo, Faculty of Engineering — Cumulative GPA: 3.95 / 4.00

Sept 2023 – Apr 2028

Waterloo, ON

SKILLS

CAD & Design: SolidWorks, Inventor, Fusion, Onshape, Blender — parametric modeling, assemblies, GD&T, rendering

Manufacturing: CNC milling, CAM toolpath generation, 3D printing (FDM), vacuum systems, O-ring seals, leak testing, mill, drill press, bandsaw

Programming & Automation: MATLAB, Python, serial communication, image & spectral analysis, hardware–software integration

Lab & Metrology: Precision measurement, XRD, ellipsometry, wet lab, optical system design, laser alignment, spectroscopy, photolithography

EXPERIENCE

Mechanical R&D Intern

Irradiant Technologies

Jan 2026 – Apr 2026

Greater Boston, MA

- Designed and CNC-machined a custom vacuum chuck for precision fixturing of delicate die structures, including full CAM toolpath generation and vacuum validation.
- Engineered and manufactured an in-house chemically resistant high-pressure die holder with asymmetric ID geometry for reliable chamber orientation.
- Modeled and rendered a two-photon lithography system supporting an SPIE Photonics West publication (Paper 13899-11).

Photolithography R&D Intern

Irradiant Technologies

Sept 2025 – Dec 2025

Greater Boston, MA

- Optimized photolithography exposure processes, improving resist performance and uniformity.
- Developed 3-axis stage characterization software for improved alignment accuracy.
- Built a full parametric CAD model of a lithography system from physical measurements.

Quantum Photonics R&D Assistant

Institute for Quantum Computing (IQC)

Jan 2025 – Apr 2025

Waterloo, ON

- Built optical assemblies for laser locking (DAVLL) and designed 3D-printed fiber routing systems.
- Automated fine-structure splitting analysis using single-photon counting instrumentation.

Quantum Photonics R&D Assistant

Institute for Quantum Computing (IQC)

May 2024 – Aug 2024

Waterloo, ON

- Increased experimental throughput by $\sim 300\times$ (174 QDs/hour) via MATLAB-based automation.
- Automated full experimental control using serial communication with stages, lasers, and spectrometers.

PROJECTS

Elevated Agitation & Thermal Wash Platform

Jan 2026 – Apr 2026

- CNC-machined elevated platforms enabling simultaneous heating and agitation, reducing solvent trapping and wash time.
- Selected materials for chemical and thermal compatibility; designed integrated transfer tools.

NWQD Operating App — Quantum Dot Control System

May 2024 – Apr 2025

- Built a MATLAB app integrating stage control, spectroscopy, imaging, and automation.
- Analyzed 7,466 quantum dots via raster scanning and automated fitting, achieving $\sim 200\times$ speedup.

Carvera Tool Wear Inspection Fixture

Jan 2026 – Apr 2026

- Designed a microscope-mounted fixture for evaluating CNC tool wear with precise positioning control.

Additional projects at mehbem.github.io

PUBLICATIONS

A Low-Cost, Open-Access Two-Photon Lithography Tool

Jan 2026

- SPIE Photonics West — Paper 13899-11

Finding the Perfect Quantum Dot for a Quantum Repeater

Jul 2025

- SPIE Photonics for Quantum 2025, Vol. 13563, 1356305.